

LESSON 305 (1.2 DUAL 1.0 INSTRUMENT)

Lesson Objective

During this lesson, the student will review aircraft control by reference to instruments. This lesson will reinforce the concept of control and performance and the fundamental skills of instrument cross-check, interpretation, and aircraft control.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
- Student questions.

Lesson Content (Ground)

Go/no-go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Flight)

Aircraft Checklist Items

Emphasize timeliness to minimize ground time!

- Instrument Cockpit Check
 - Student Tells, Student Does.
 - Have the student read each item from the checklist out loud.
 - Demonstrate to the student what needs to be done for each item.
 - Discipline the student to set bugs, bearing pointers, and any auxiliary features.
 - The autopilot check can be conducted now:
 - Engage autopilot.
 - Test roll: left and right.
 - Test pitch: up and down (VS mode).
 - Ensure autopilot can be overridden.

- Check the red disconnect button.
- Electric trim test:
 - Check the motion of the trim wheel.
 - Check the S-TEC trim annunciators.
 - Each half-switch should not move the trim wheel.
 - The disconnect switch should stop the wheel.
- Before taxi check:
 - Student Tells, Student Does.
 - Airspeed indicator (PFD and standby): zero.
 - Attitude indicator (PFD): pitch/bank deviations +/- 5 degrees.
 - Standby attitude indicator: standby battery check and adjust the miniature aircraft.
 - Altimeter (PFD and standby): +/- 75 feet of station elevation.
 - Vertical speed indicator (PFD): reads 0 or use current indication.
 - Trim coordinator: wings level or no vector when still; inclinometer centered when still; skids when turning.
 - HSI: matches compass and turns appropriately.
 - Magnetic compass: indicates heading properly, no bubbles, card present, and swings freely.

Departure

- Follow tower's instructions for departure.
- The student must put the hood or foggles on when safe.
- Climb out:
 - Student Does, Instructor Evaluates.
 - Give vectors to the practice area.
 - Maintain V_Y .
 - Conduct the 1,000ft check.
 - *IP deviates from cruise climb:*
 - Request nonstandard constant rate and airspeed climbs.
 - Level off at a given altitude.

Full Panel Maneuvers

- Pattern F:
 - Instructor Tells, Student Does.
 - Instructor keeps time.
 - Discuss airspeed changes and bank.
 - Recommend known power settings.
 - Note time – recover at Training Cruise.
- Steep turns:
 - Student Does, Instructor Evaluates.
 - Conduct the pre-maneuver checklist.

- Coach fast scan and fine touch on the controls.
- Recommend added power and trim.
- Ensure that the rollout is timed appropriately ($\frac{1}{2}$ bank angle before).
- Repeat in the opposite direction.
- Recognition and recovery from unusual attitudes:
 - Student Does, Instructor Evaluates.
 - Conduct the pre-maneuver checklist.
 - Disorient the student - use slips and skids.
 - Prompt recovery.
 - On the last unusual attitude, dim the PFD.
 - Prompt recovery (ensure that the student is now using the standby instruments).
 - Resume partial panel straight and level flight.

Partial Panel Maneuvers

- Pattern E:
 - Instructor Tells, Student Does.
 - The instructor will call the turns.
 - For this pattern, have the student time turns to headings.
 - Coach the student on changing the angle of bank when airspeeds change to maintain standard rate.
 - Verify the accuracy of each turn.

Arrival

- Vector the students to a reporting point.
- Prompt the descent and arrival checklist.
- Simulated ASR approach:
 - Instructor Tells, Student Does.
 - Assign headings.
 - Assign constant rate and airspeed descents.
 - Reduce speed according to the location in the pattern.
 - Go visual and allow the student to land.

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 306.
- Assign Homework:
 - FIT procedures for VOR Intercepting and Tracking.

- FIT procedures for Intercepting and Tracking DME Arcs from radials and radar vectors.
- Instrument Flying Handbook:
 - Chapter 09 - Navigation Systems.

Completion Standards

This lesson will be complete when the student can, with minimal instructor assistance:

1. Control the aircraft to maintain altitude within + 200 feet, headings + 15 degrees and airspeeds + 15 knots.

Pattern E



